

Approximation of Mutual Information in Random Walk Voltage Models

Problem Setting

Consider a graph with n nodes. We designate k source nodes $S_1, \dots, S_k$ and a single sink node T . For each source S_i , define the voltage function $V_i(u)$ on nodes $u \in \{1, \dots, n\}$ via the Dirichlet problem:

$$\begin{cases} V_i(S_i) = 1, \\ V_i(T) = 0, \\ \Delta V_i(u) = 0 \text{ for } u \notin \{S_i, T\}. \end{cases}$$

Each $V_i(u)$ represents the probability that a random walk starting at node u reaches source S_i before the sink T .

Suppose N independent particles are released from node u . For each source S_i , let $m_i(u)$ be the number of particles that hit S_i before T . Then:

$$m_i(u) \sim \text{Binomial}(N, V_i(u)).$$

Define the k -dimensional random vector at node u as:

$$A(u) = (m_1(u), m_2(u), \dots, m_k(u)).$$

We introduce a random variable B that is uniformly distributed over the n nodes:

$$B \sim \text{Uniform}\{1, 2, \dots, n\}.$$

We are interested in the mutual information $I(A; B)$, where $A = A(B)$.

Conditional Distribution of A

Conditioned on $B = u$, the components of A are independent:

$$A \mid B = u \sim \bigotimes_{i=1}^k \text{Binomial}(N, V_i(u)).$$

For large N , each binomial distribution can be approximated by a Gaussian:

$$\text{Binomial}(N, p) \approx \mathcal{N}(Np, Np(1-p)).$$

Therefore,

$$A \mid B = u \approx \mathcal{N}(\mu(u), \Sigma(u)),$$

where

$$\begin{aligned} \mu_i(u) &= NV_i(u), \\ \Sigma(u) &= \text{diag}(NV_1(u)(1-V_1(u)), \dots, NV_k(u)(1-V_k(u))). \end{aligned}$$

Mutual Information Formula

The mutual information is:

$$I(A; B) = H(A) - H(A|B).$$

Conditional Entropy $H(A|B)$

Given $B = u$, A is approximately Gaussian with covariance $\Sigma(u)$. The differential entropy of a multivariate Gaussian is:

$$H(A|B = u) = \frac{k}{2} \log(2\pi e) + \frac{1}{2} \log \det \Sigma(u).$$

Averaging over uniform B :

$$H(A|B) = \frac{1}{n} \sum_{u=1}^n H(A|B = u).$$

Marginal Entropy $H(A)$

The marginal distribution of A is a mixture of Gaussians:

$$P(A) = \frac{1}{n} \sum_{u=1}^n \mathcal{N}(\mu(u), \Sigma(u)).$$

Its entropy $H(A)$ has no closed-form expression. We approximate it via Monte Carlo sampling:

- Sample B uniformly from $\{1, \dots, n\}$.
- Sample $A | B$ from $\mathcal{N}(\mu(B), \Sigma(B))$.
- Estimate $p(A)$ using the mixture density.
- Approximate $H(A) \approx -\frac{1}{M} \sum_{j=1}^M \log p(A^{(j)})$.

Monte Carlo Approximation Procedure

1. For each node u , compute:

$$\mu(u) = N \cdot V(u), \quad \Sigma(u) = \text{diag}(NV_i(u)(1 - V_i(u))).$$

2. Compute:

$$H(A|B) = \frac{1}{n} \sum_{u=1}^n \frac{k}{2} \log(2\pi e) + \frac{1}{2} \sum_{i=1}^k \log(NV_i(u)(1 - V_i(u))).$$

3. Draw M samples:

$$\begin{aligned} B^{(j)} &\sim \text{Uniform}\{1, \dots, n\}, \\ A^{(j)} &\sim \mathcal{N}(\mu(B^{(j)}), \Sigma(B^{(j)})). \end{aligned}$$

4. For each sample $A^{(j)}$, compute:

$$p(A^{(j)}) \approx \frac{1}{n} \sum_{u=1}^n \mathcal{N}(A^{(j)}; \mu(u), \Sigma(u)).$$

5. Approximate:

$$H(A) \approx -\frac{1}{M} \sum_{j=1}^M \log p(A^{(j)}).$$

6. Finally:

$$I(A; B) \approx H(A) - H(A|B).$$

Interpretation

- For large N , the binomial distributions are sharply peaked, so the means of the Gaussians separate well.
- If the voltage vectors $V(u)$ are well-separated across nodes, $I(A; B)$ approaches $\log n$.
- If the voltage vectors are similar across nodes, $I(A; B)$ is small.

This approximation provides a practical way to quantify how well observing the particle count vector A can identify the underlying node B in large- N regimes.